How does laplacian differential privacy injection affect the convergence, accuracy, and performance of Convolutional Neural Networks in the context of financial data using empirical data from published papers?

1. Which tool did you use?
We used Google Scholar to find peer-reviewed papers and research articles on differential privacy and convolutional neural networks.

2. What keywords or Boolean operators did you apply?
We used a combination of keywords and Boolean connectors:

- “differential privacy” AND “convolutional neural network” AND “financial data”
- “Laplacian noise” AND “DP-SGD” AND accuracy
- “differential privacy” AND “CNN” AND inference latency OR energy efficiency

These helped narrow the search to papers that specifically address privacy techniques, CNN performance, and financial data applications.

3. What does this tool do?
Google Scholar indexes academic research including journal papers, conference proceedings, and theses. It provides links to original sources and often free PDFs. It also gives citation counts, which help identify influential work.

4. How did you use it to answer your research question?
We searched for empirical studies where CNNs were trained with differential privacy (particularly with Laplacian noise) and reviewed their results. From these papers, we extracted information on how DP affected training speed, accuracy, inference performance, and energy efficiency. We then compared findings across studies, with extra focus on financial datasets.

5. One strength of the tool

Google Scholar is free, easy to use, and provides access to a very large pool of scholarly content across multiple disciplines. It also ranks results by relevance and citation count, which helps in quickly finding high-quality sources.

6. One limitation of the tool

Not all papers are freely accessible through Google Scholar. Many are behind paywalls, which means sometimes only the abstract is available unless you have institutional access. It also mixes in some less relevant or non-peer-reviewed sources if you are not careful with filtering.